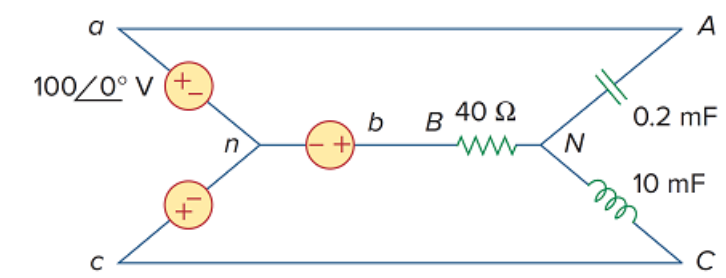


Problem

The source in Fig. 12.65 is balanced and exhibits a positive phase sequence. If $f = 60$ Hz, use *PSpice* or *MultiSim* to find $\mathbf{V}_{AN}$, $\mathbf{V}_{BN}$, and $\mathbf{V}_{CN}$.

Figure 12.65



Step-by-step solution

Step 1 of 2

The schematics is shown below. In the AC sweep box, we set total pts = 1, Start frequency = 60 & End frequency = 60.

After simulation, we obtain an output file which includes,

Frequency	VM(1)	VP(1)
6.000 E+01	2.206 E+02	-3.456 E+01
Frequency	VM(2)	VP(2)
6.000 E+01	2.141 E+02	-8.149 E+01
Frequency	VM(3)	VP(3)
6.000 E+01	4.991 E+01	-5.059 E+01

Step 2 of 2

i.e.,

$$V_{AN} = 220 \angle -34.56^\circ$$

$$V_{BN} = 214.1 \angle -81.98^\circ$$

$$V_{CN} = 49.91 \angle -50.85^\circ$$

